RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Tuesday, 15 September 2026, 11:25 AM
State	Finished
Completed on	Tuesday, 15 September 2026, 11:32 AM
Time taken	7 mins 23 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int twoSum(int arr[], int low, int high, int x, int *e1, int *e2) {
    if (low >= high)
        return 0;

    int mid = (low + high) / 2;

    if (twoSum(arr, low, mid, x, e1, e2))
        return 1;

    if (twoSum(arr, mid + 1, high, x, e1, e2))
        return 1;
    int i = mid, j = mid + 1;
    while (i >= low && j <= high) {
        int sum = arr[i] + arr[j];
        if (sum == x) {
            *e1 = arr[i];
            *e2 = arr[j];
            return 1;
        } else if (sum < x) {
            j++;
        } else {
            i--;
        }
    }

    return 0;
}

int main() {
    int n;
    scanf("%d", &n);

    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    int x;
    scanf("%d", &x);

    int e1, e2;
    if (twoSum(arr, 0, n - 1, x, &e1, &e2)) {
        printf("%d\n", e1);
        printf("%d\n", e2);
    } else {
        printf("No\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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